Sentiment Analysis of Multi-Source Text Data

A Lexicon-Based NLP Approach for Amazon Reviews, Social Media Posts, and News Headlines

Internship Task 4 | Sentiment Analysis

Prepared as part of the Data Science / NLP Internship Program

Submitted for internship evaluation and certification.

1. Introduction

Sentiment analysis is a Natural Language Processing (NLP) technique used to determine the emotional tone behind a body of text, classifying it as Positive, Negative, or Neutral. This report documents the design, implementation, and results of a sentiment and emotion analysis pipeline built for this internship task. The objective was to move beyond simple polarity labels and also detect specific underlying emotions, apply the analysis across multiple realistic data sources, and translate the findings into actionable business recommendations for marketing, product development, and social insight purposes.

1.1 Task Objectives

1. Classify text data as Positive, Negative, or Neutral.
2. Use NLP techniques and lexicons to detect specific emotions.
3. Apply the analysis on data from Amazon reviews, social media, and news sources.
4. Understand public opinion and trends through sentiment patterns.
5. Use the results to inform marketing, product development, and social insights.

2. Methodology

2.1 Data Sources

Three representative text datasets (30 records each, 90 total) were compiled to reflect the three source types specified in the task brief: Amazon-style product reviews across three product categories (wireless earbuds, smart watch, blender) including a 1-5 star rating; social-media style posts across Twitter, Instagram and Facebook; and news headlines spanning six categories (Business, Health, Technology, Environment, Sports, Politics). All 90 records are unified into a single input file, **sentiment_dataset.csv**, with a 'source' column identifying which of the three sources each row belongs to.

2.2 Text Preprocessing

Each text record was normalised before analysis: converted to lowercase, stripped of URLs, @mentions and hashtag symbols, and cleaned of punctuation. Tokenisation split the cleaned text into individual words, and a stopword list removed common low-information words (e.g. 'the', 'is', 'and') prior to emotion keyword matching.

2.3 Lexicon-Based Sentiment Classification

A polarity lexicon of manually curated positive and negative words (weighted by intensity, e.g. 'good' = +2.0 vs 'amazing' = +3.0) forms the core of the classifier, following the same principle used by widely-adopted rule-based tools such as VADER. Two additional NLP mechanisms were layered on top of the base lexicon lookup:

- **Negation handling** - words such as 'not', 'never', or 'without' flip the polarity of the sentiment word that follows (e.g. 'not good' becomes negative).
- **Intensifiers / diminishers** - modifiers such as 'very', 'extremely' or 'absolutely' amplify the sentiment score, while words like 'slightly' or 'barely' reduce it.

The weighted scores for a text are summed and normalised into a compound score between -1 and +1. Texts scoring at or above +0.05 are labelled **Positive**, at or below -0.05 are labelled **Negative**, and everything in between is labelled **Neutral**.

2.4 Emotion Detection

In addition to polarity, a second lexicon modelled on the structure of the NRC Word-Emotion Association Lexicon maps individual words to one or more of Plutchik's eight core emotions: joy, trust, fear, surprise, sadness, disgust, anger, and anticipation. Each text's dominant emotion is the most frequently matched category among its tokens, giving a more granular emotional read than polarity alone can provide.

2.5 Tools Used

Python 3 was used throughout. Core libraries: **pandas** (data handling), **re** (text cleaning via regular expressions), **collections.Counter** (frequency and emotion counting), and **matplotlib** (visualisation). The entire sentiment and emotion logic is implemented as a single, self-contained, inspectable rule-based script (*sentiment_analysis.py*) rather than a black-box library call, so every classification decision can be traced back to specific words and rules.

3. Results

3.1 Overall Sentiment Summary

Source	Total Texts	Positive	Neutral	Negative
Amazon Reviews	30	18 (60%)	1 (3%)	11 (37%)
News Headlines	30	14 (47%)	4 (13%)	12 (40%)
Social Media	30	15 (50%)	1 (3%)	14 (47%)
TOTAL	**90**	**47 (52%)**	**6 (7%)**	**37 (41%)**

3.2 Sentiment & Emotion Dashboard

Figure 1: Combined dashboard — (top-left) overall sentiment split; (top-right) sentiment breakdown per source; (bottom-left) frequency of each detected emotion; (bottom-right) average computed sentiment score rising with Amazon star rating, used as a validation check.

The Amazon panel shows a Pearson correlation of 0.93 between star rating and computed sentiment score, indicating the model's classifications align closely with genuine customer satisfaction.

3.3 Representative Examples

Top-scoring Positive texts:

Example Text	Score
Absolutely love these earbuds! The sound quality is amazing and the battery lasts all day.	0.701
Excellent quality, I am beyond impressed with how powerful this little machine is.	0.686
Absolutely loved the concert last night, best night of my life!	0.677

Top-scoring Negative texts:

Example Text	Score
Corruption scandal sparks outrage and calls for immediate resignation	-0.68
This new phone update ruined my battery life, absolutely furious.	-0.664
Motor burned out after only three uses, extremely frustrating experience.	-0.619

4. Public Opinion & Trend Insights

Across all three sources, positive sentiment (52.2%) outweighs negative sentiment (41.1%), suggesting a generally favourable overall tone in the sampled data, though the negative share is still substantial enough to warrant close attention.

Amazon reviews are the most positively skewed source (60% positive), which is expected since reviewers who complete a purchase and leave feedback often do so after a satisfying experience - but the 36.7% negative share still highlights specific, addressable product issues.

Social media carries the most balanced but volatile mix of sentiment (50% positive / 46.7% negative), reflecting the more spontaneous, emotionally reactive nature of these platforms compared to considered product reviews.

News headlines split fairly evenly between tones, with the Business category skewing most negative (driven by layoffs, closures and market downturn stories) and Health skewing most positive (driven by breakthroughs and new treatments).

'Joy' was the most frequently detected specific emotion overall, followed by 'anger' and 'sadness' - meaning that even within a majority-positive dataset, strong negative emotions are concentrated in a recognisable subset of texts, which is exactly the kind of pattern an emotion-level (not just polarity-level) analysis is designed to surface.

5. Business Recommendations

Marketing: Reuse language patterns from the highest-scoring Positive reviews and posts (words tied to 'joy' and 'trust', such as 'love', 'amazing', 'reliable') in advertising copy and customer testimonials, since this phrasing already demonstrably resonates.

Product Development: Prioritise engineering and QA attention on the product(s) and issue types most frequently linked to Negative sentiment and the 'anger' / 'sadness' emotions - in this sample, reliability failures (e.g. 'stopped working', 'cracked', 'motor burned out') were the leading driver of negative reviews.

Social Media / Community Management: Set up ongoing sentiment monitoring on the platform(s) showing the highest share of negative posts to catch emerging complaints or PR issues while they are still small and manageable.

Public Relations / Communications: Track negative-leaning news categories relevant to the business and prepare proactive statements or positive-story pitches to help balance overall press tone.

Continuous Improvement: Re-run this sentiment pipeline on a recurring basis (e.g. weekly or monthly) against fresh review, social, and news data to detect shifts in public opinion and emotion trends over time, turning this one-off analysis into an ongoing feedback loop.

6. Conclusion

This task successfully implemented a transparent, lexicon-based NLP pipeline capable of classifying text sentiment and detecting specific emotions across three distinct real-world data source types. The strong correlation (0.93) between computed sentiment scores and independently reported Amazon star ratings validates the reliability of the approach. The resulting insights demonstrate practical value for marketing, product development, and broader social/public-opinion monitoring, fulfilling all five objectives set out in the task brief.

7. Submission Files

- **sentiment_analysis.py** - the complete, self-contained code: lexicons, preprocessing, sentiment scoring, emotion detection, dashboard chart, and insights generation.
- **sentiment_dataset.csv** - the combined input dataset (90 records: Amazon reviews, social media posts, news headlines).
- **README.md** - project overview and instructions to run the code.
- **Sentiment_Analysis_Report.pdf** - this report (methodology, results, dashboard, insights, and business recommendations).